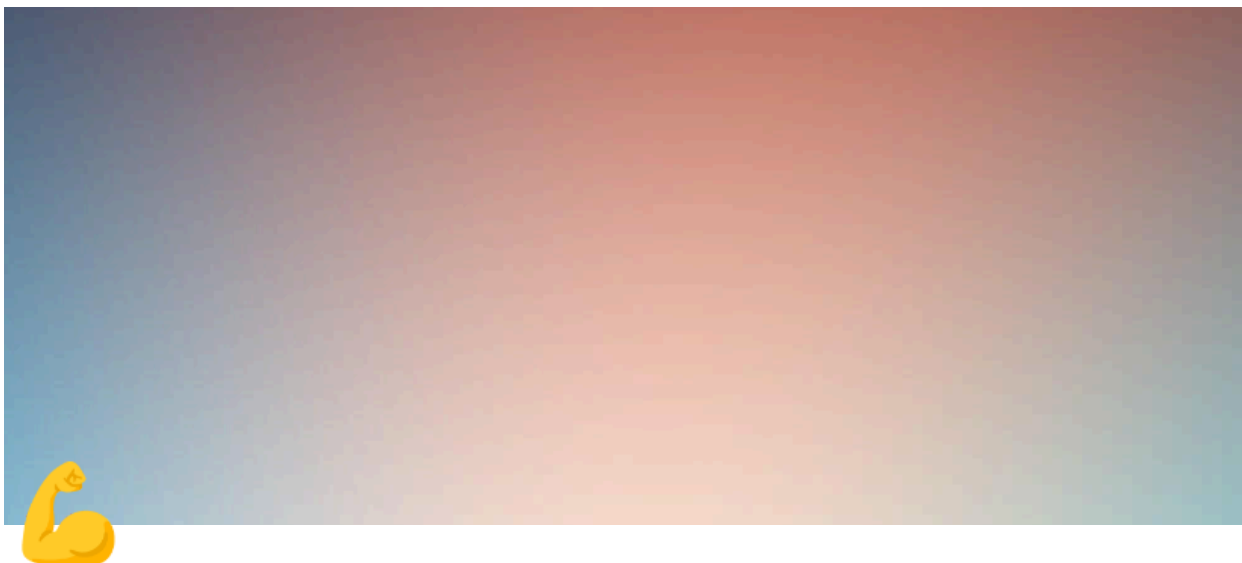




From Blind to Aware: Building Your Own Security Operations Center Using Free Tools

Introduction

What is **Security Operations Center (SOC)**?

A **Security Operations Center (SOC)** is a centralized system that collects, analyzes, and monitors security events across devices to detect and respond to threats in real time. Many organizations rely on SOC's to identify and respond to cyberattacks as they happen.

The Scenario

It is 3:17 on a Tuesday morning. You are fast asleep. Your laptop is on the desk, closed, but still powered on while someone on the other side of the world is wide awake, and they are trying to break into your computer. They are trying passwords, running commands, and probing for weaknesses, looking for any way in.

Meet Alex. Alex runs a small graphic design business from home, with three computers and a drive full of client work. Like most people, Alex assumes

everything is fine. But last Tuesday at 3 AM, someone attempted to log into one of those machines and Alex has no idea. Alex's story is not unique. It is happening every day, to individuals and organizations alike.

The Problem

If you were Alex, would you know if someone was in your system? No. You would wake up, check your messages, and go about your daily activities.. without realizing anything had happened. Even if an attacker gained access, the compromise could persist silently in the background without your knowledge.

According to IBM Security's "Cost of a Data Breach Report", the average time between a hacker getting into a system and someone discovering them is 207 days. That is nearly seven months of them watching you type your passwords. Seven months of them reading your emails. Seven months of them slowly stealing everything of value.

And the sad part is that most victims only find out when it is too late. When their bank account is empty, their identity is stolen or when their important documents are missing.

The Solution

The solution is to build your own **mini Security Operations Center (SOC)**.

This involves setting up a central system that continuously monitors your devices, collects logs, and alerts you in real time when suspicious activity occurs. Instead of being blind to attacks, you gain full visibility into what is happening across your systems. While organizations invest millions in SOC infrastructure and dedicated security teams, it is possible to build a functional version using free and open-source tools, right from your own environment.

By the end of this project, you will have built a working mini-SOC capable of detecting and analyzing security events like a real-world security analyst.

What You Will Need

Tool	Purpose	What it does
WAZUH	SIEM (Security Information and Event Management)	Open-source, enterprise-grade, used by real companies. We can access the dashboard via the browser

Tool	Purpose	What it does
SOC Server (Ubuntu)	Our SOC platform	Central server that collects logs, analyzes data, hosts dashboard
VirtualBox	Run all our VMs	Free, works on Windows/Mac/Linux
Wazuh Agent (Host-Windows & Windows VM)	Installed on monitored devices	Sends logs to our SOC server
Kali VM	Attacker Machine	Used to simulate attacks to other VMs

Table 1: Tools Used in the Home SOC Lab

The good thing about the above listed tools is the fact that they are all free.

Part 1. The Architecture

This is a quick summary of what our SOC will look like

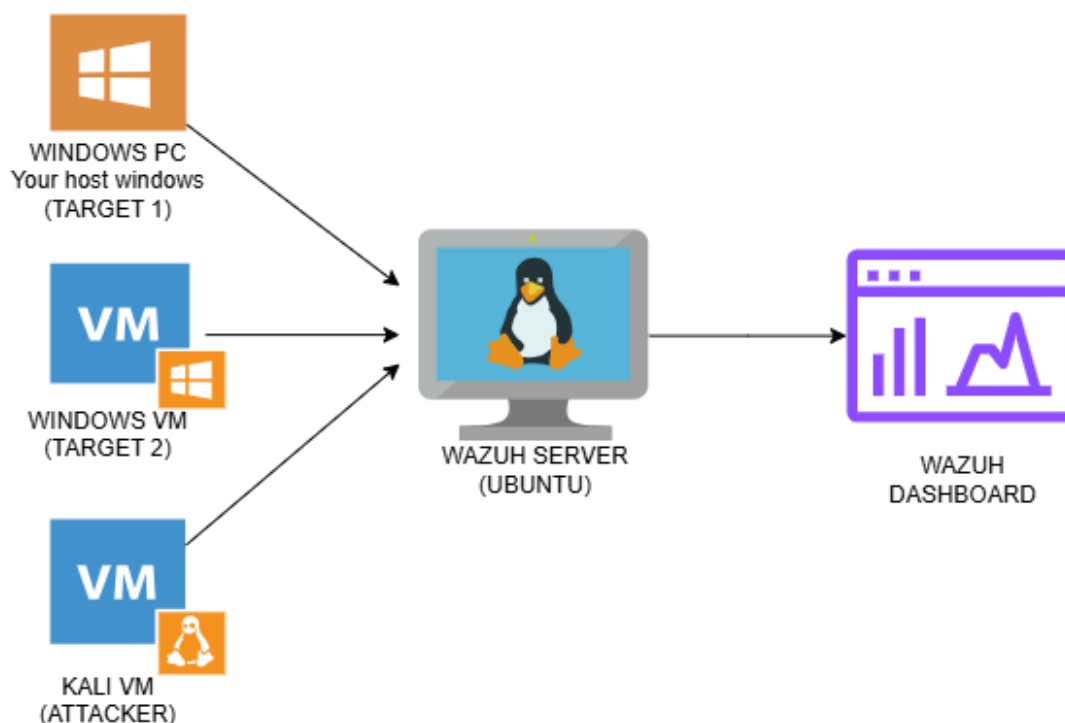


Figure 1: Network Architecture of the Home SOC Lab Environment

Here's what is actually happening:

- Your main computer and Windows VM send logs to the Ubuntu server
- The Ubuntu server runs Wazuh, which analyzes everything in real-time
- The Wazuh dashboard lets you see alerts in your browser
- Kali Linux is your attacker which you will use it to test if your SOC work

Part 2: Building Your SOC

Step 2.1: Install VirtualBox

Go to <https://www.virtualbox.org>, download the version for your operating system, and run the installer. Click *Next* on everything, it is very straightforward.

Step 2.2: Create your SOC server (Ubuntu)

a. Download Ubuntu

Go to the official Ubuntu site: <https://ubuntu.com/download/desktop> and download the latest LTS version (Ubuntu 24.04 LTS at the time of this writing). "LTS" means Long Term Support, it is stable and well-tested.

b. Create the virtual machine

Open VirtualBox, click **New**, and set it up like this:

- Name: **UbuntuSOC** (or whatever you prefer)
- Type: **Linux**
- Version: **Ubuntu (64-bit)**
- Memory: 4096 MB (this matters for smooth performance)
- Hard disk: Create a virtual hard disk → VDI → Dynamically allocated → 20 GB minimum

c. Adjust VM settings before starting

- System → Processor: 2 CPUs minimum
- Display → Video Memory: 128 MB
- 3D Acceleration: OFF (not needed)
- Storage → Click the empty optical drive, then the disk icon, and select your Ubuntu ISO

- Network → Set to *Bridged Adapter* (this puts your VM on the same network as your main computer)
- Shared Clipboard: Set to *Bidirectional* (so you can copy commands between windows)

d. Install Ubuntu

Start the VM. Select *Install Ubuntu* follow the prompts (language, keyboard, create a user), and let it finish. When it is done, reboot.

e. Find your server's IP

Open the terminal (the black icon) and type *ip a* to find its IP

```
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group def
ault qlen 1000
    link/ether 08:00:27:d0:d8:4f brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8
        valid_lft 371sec preferred_lft 371sec
    inet6 fe80::222e:620:caae:d364/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 2.1: Finding Your Ubuntu Server's IP Address

Then update and upgrade your ubuntu to be ready for the exercise. Type the following command simultaneously to do so; *sudo apt update && sudo apt upgrade -y*. This fetches the latest security patches and software updates.

Step 2.3: Set Up Your Attacker Machine (Kali)

a. Download Kali Linux

Go to <https://www.kali.org/get-kali/>, look for the Virtual Machines section. Download the VirtualBox version (it's pre-built, so you don't have to install anything)

b. Import Kali and Configure its Network

In VirtualBox, click *File* → *Import Appliance*. Find the *.ova* file you downloaded, click through, and let it import. Then right-click *Kali VM* → *Settings* → *Network*. Set it to *Bridged Adapter*, same as your Ubuntu VM. It is important for your Kali machine to be on the same network as your SOC server so they can talk to each other.

d. Start Kali and Find Its IP

- Start the Kali VM and login with:

Username: `kali`

Password: `kali`

Open a terminal (the black icon at the top) and type; `ip a` to find its IP

```
(kali㉿kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:b4:a1:05 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.4/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 437sec preferred_lft 437sec
    inet6 fe80::d699:a8b4:87b1:3426/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:d0:58:74 brd ff:ff:ff:ff:ff:ff
    inet 192.168.137.64/24 brd 192.168.137.255 scope global dynamic noprefixroute eth1
        valid_lft 604640sec preferred_lft 604640sec
    inet6 fe80::c13f:3373:5015:2df9/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Figure 2.2: Finding Your Kali VM's IP Address

Step 2.4: Create Your Target Machine (Windows VM)

a. Download Windows 10/11

Microsoft provides trial ISOs. Download one, or if you have a Windows license, use that.

b. Create the VM

- Name: `WindowsTarget` (or anything you choose)
- Type: `Microsoft Windows`
- Version: Your Windows version
- Memory: 4096 MB
- Hard disk: 40 GB dynamically allocated
- Network: Bridged Adapter (keeping everything on the same network)

c. Install Windows

Boot from the ISO, go through installation, and create a local account (you don't need to activate it for this lab). Check your ip by using the terminal in powershell (run as administrator)

```
PS C:\Windows\system32> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : mshome.net
    Link-local IPv6 Address . . . . . : fe80::1330:a19a:47a3:1590%3
    IPv4 Address. . . . . : 10.0.2.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.0.2.1

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::85a1:af5:12ee:4bc0%5
    IPv4 Address. . . . . : 192.168.56.103
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

Figure 2.3: Finding Your Windows VM's IP Address

Step 2.5: Set up your host Windows

First check your ip by using the terminal in powershell (run as administrator)

```
PS C:\Windows\system32> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::50db:c716:3df1:7103%9
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . :
```

Figure 2.4: Finding Your Host-Windows's IP Address

Part 3: Installing Wazuh on Your SOC Server (Ubuntu)

Now we'll install the actual security platform which is Wazuh.

3.1: Download and Run the Wazuh Installer

From your Ubuntu terminal, use root to download and run the Wazuh Installer by typing `sudo su` and enter followed by the command below;

```
# Install curl first
sudo apt update && sudo apt upgrade -y
sudo apt install curl -y
curl -s0 https://packages.wazuh.com/4.12/wazuh-install.sh
curl -s0 https://packages.wazuh.com/4.12/config.yml
chmod +x wazuh-install.sh
sudo ./wazuh-install.sh --generate-config-files
```

The installer begins generating configuration files

3.3: Complete the Installation

Run the installation assistant in order to the install Wazuh Indexer, Server, and Dashboard

```
sudo ./wazuh-install.sh -a -i
```

-a: Installs indexer, server, and dashboard together

-i: Bypasses system compatibility check (needed for Ubuntu 24.04 which is what I use)

3.4: Get Your Admin Credentials

After installation, the script displays your login information. Copy the login information somewhere safe because you would be needing it many times.

```
13/03/2026 02:33:00 INFO: Wazuh dashboard web application not yet initialized. Waiting...
13/03/2026 02:33:17 INFO: Wazuh dashboard web application not yet initialized. Waiting...
13/03/2026 02:33:32 INFO: Wazuh dashboard web application initialized.
13/03/2026 02:33:32 INFO: --- Summary ---
13/03/2026 02:33:32 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: LtebF80J*1hUoah53VHwaVsFUitXJMtm
13/03/2026 02:33:32 INFO: Installation finished.
```


Figure 3.1: Wazuh Installation Output with Admin Credentials.

Step 3.6: Access the Wazuh Dashboard

1. From your main computer, open a web browser
 - a. Go to: `https://192.168.56.102` (use your SOC server's IP)
2. You will see a security warning - click *Advanced* then *Proceed*

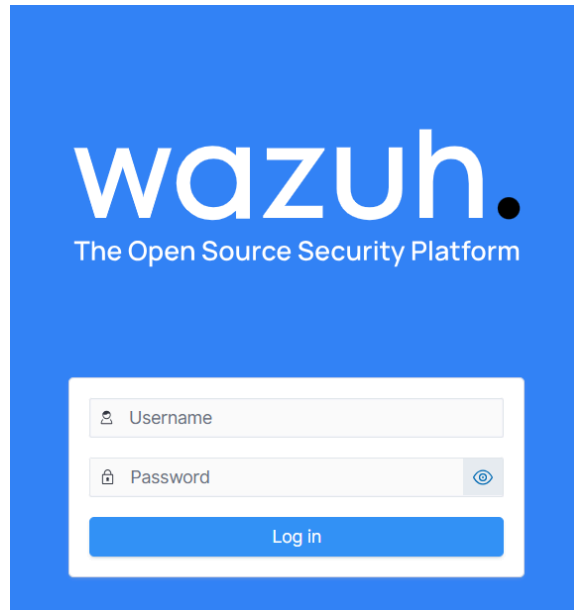


Figure 3.2: Navigating the SSL Certificate Warning to Reach the Wazuh Dashboard

Log in with the username and password provided in your ubuntu terminal. Now, your SOC is live.

Part 4: Connecting Devices to your SOC (Wazuh)

A security center with no data is like a security camera that is not plugged in. We need to connect devices to our SOC. We'll monitor three machines:

1. Your main computer (I used Windows)
2. A target VM (I used Windows VM)
3. Kali Linux (to simulate attacks)

4.1: Install Wazuh Agent on Windows VM

a. Download the agent:

- On your Windows VM, open browser and go to: <https://packages.wazuh.com/4.x/windows/wazuh-agent-4.7.2-1.msi>
- Download the file, run the installer by double-clicking the downloaded file.
- When prompted for Wazuh server address, enter your Ubuntu sever's IP ([192.168.56.102](#)) and leave other settings default and complete installation. If by any chance, your wazuh agent setup shows a pop-up like this;

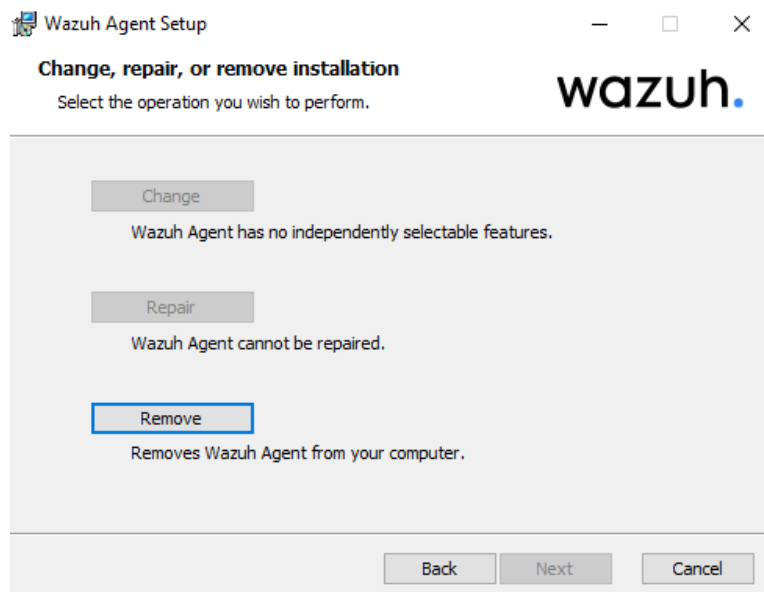
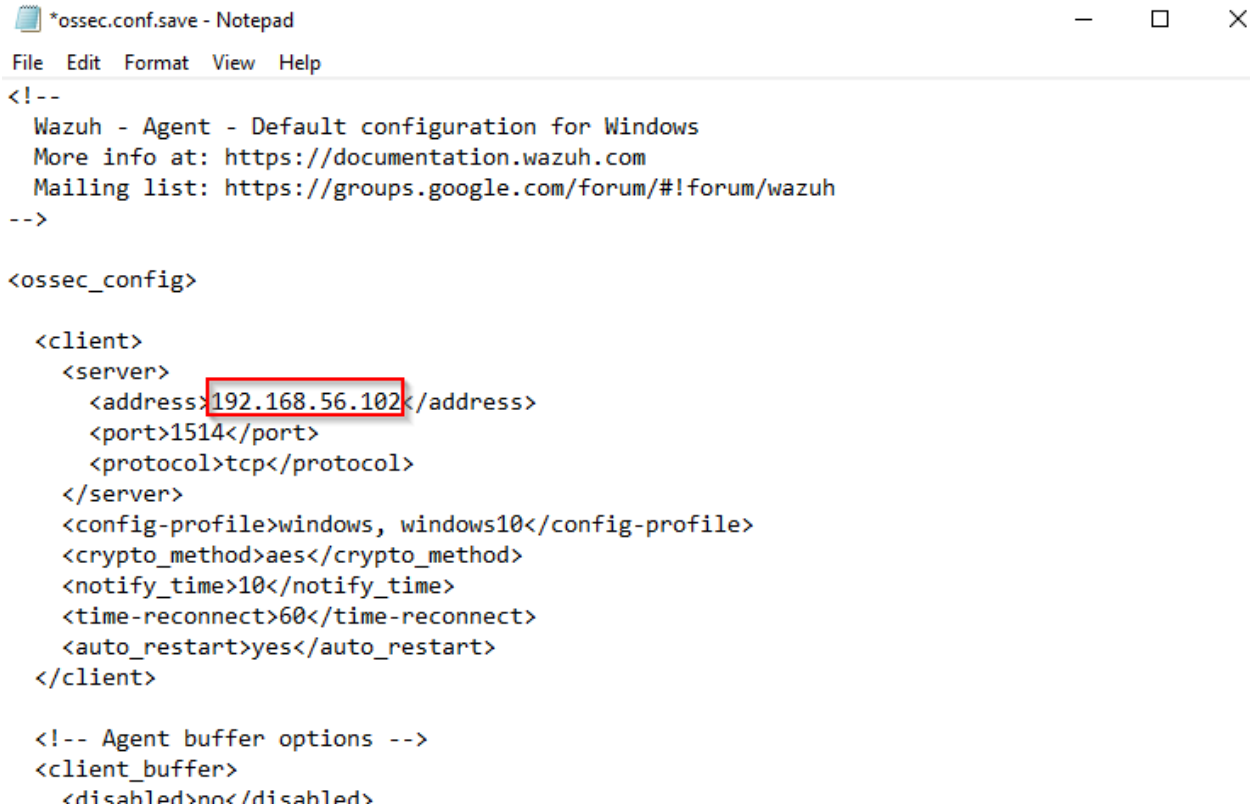


Figure 4.1: Wazuh Agent Prompt Indicating Automatic Configuration

It just means that your wazuh agent has been automatically set up, inorder words, do not click anything.

b. Configure the agent.

- Open Notepad as Administrator, click *Start* and type *notepad*, right-click it and "*Run as administrator*"
- Inside Notepad, Click *File* → *Open* → *Program files (x86)* → *ossec agent* → *ossec.conf*. Remember to change file type from "*Text Documents (*.txt)*" to "*All Files (*.*)*"
- Find the line that says `<address>MANAGER_IP</address>` Replace `MANAGER_IP` with your Ubuntu server's IP then save and close.



```
*ossec.conf.save - Notepad
File Edit Format View Help
<!--
Wazuh - Agent - Default configuration for Windows
More info at: https://documentation.wazuh.com
Mailing list: https://groups.google.com/forum/#!forum/wazuh
-->

<ossec_config>

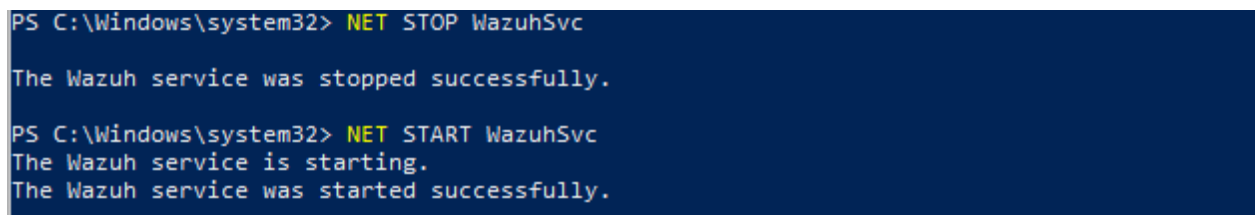
  <client>
    <server>
      <address>192.168.56.102</address>
      <port>1514</port>
      <protocol>tcp</protocol>
    </server>
    <config-profile>windows, windows10</config-profile>
    <crypto_method>aes</crypto_method>
    <notify_time>10</notify_time>
    <time-reconnect>60</time-reconnect>
    <auto_restart>yes</auto_restart>
  </client>

  <!-- Agent buffer options -->
  <client_buffer>
    <disabled>no</disabled>
```

Figure 4.1.2 Editing ossec.conf to Point to the SOC Server

- Restart agent by opening *PowerShell* as Administrator and type this;

```
NET STOP WazuhSvc
NET START WazuhSvc
```



```
PS C:\Windows\system32> NET STOP WazuhSvc
The Wazuh service was stopped successfully.

PS C:\Windows\system32> NET START WazuhSvc
The Wazuh service is starting.
The Wazuh service was started successfully.
```

Figure 4.1.3: Restarting the Wazuh Agent to Apply Changes

Step 4.2: Extract your key from the SOC server

Back on your Ubuntu server, you need to generate a key for this agent. Run *sudo /var/ossec/bin/manage_agents* to add the agent and extract the key

```
root@ubuntu-VirtualBox:/home/ubuntu# sudo /var/ossec/bin/manage_agents

*****
* Wazuh v4.12.0 Agent manager.          *
* The following options are available: *
*****
(A)dd an agent (A).
(E)xtract key for an agent (E).
(L)ist already added agents (L).
(R)emove an agent (R).
(Q)uit.
Choose your action: A,E,L,R or Q: A
```

Figure 4.2: Adding a New Agent and Generating Its Authentication Key

Inside the menu, follow the prompt. Press A then enter:

1. Type **A** and press *Enter* to add an agent (e.g windows-agent), then enter the IP address of your Windows VM
2. Type **E** and press *Enter* to extract the key
3. Enter the agent ID and copy the long key that appears

Step 4.3: Import key from the server on your Windows VM

- a. Open *PowerShell* as Administrator and run this command;

```
& "C:\Program Files (x86)\ossec-agent\manage_agents.exe"
```

Inside the menu, Press I and press *Enter* to import a key, then paste the key you copied from Ubuntu and follow the prompt.

```
PS C:\Windows\system32> & "C:\Program Files (x86)\ossec-agent\manage_agents.exe"

*****
* Wazuh v4.7.2 Agent manager.          *
* The following options are available: *
*****
(I)mport key from the server (I).
(Q)uit.
Choose your action: I or Q: I

* Provide the Key generated by the server.
* The best approach is to cut and paste it.
*** OBS: Do not include spaces or new lines.

Paste it here (or 'q' to quit): MDAxIHdpbmRvd3MtYldlbnQgMTkyLjE2OC41Ni4xMDMgMzNjYjgyNTk8YzU4ZTY4ZmVhNDgyMzdkOTIyNjQzMTRjZGU1MTA5MmMwMjRlNzZlZW
I3MDk5NGMyOGZlMTc2MQ==

Agent information:
  ID:001
  Name:windows-agHnt
  IP Address:192.168.56.103

Confirm adding it?(y/n): Y
Added.
```

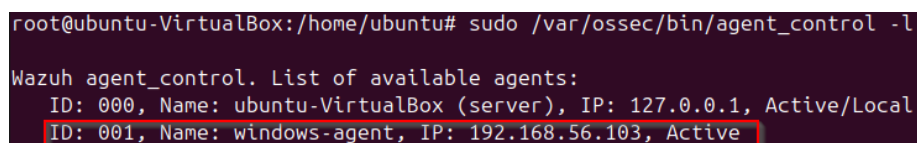
Figure 4.3: Importing the Agent Key on Windows to Complete Registration

Then restart agent (Windows) again using;

```
NET STOP WazuhSvc
NET START WazuhSvc
```

STEP 4.4 - Verify connection on the SOC server (Ubuntu)

- a. On Ubuntu, run `sudo /var/ossec/bin/agent_control -l` to verify the connection.

A terminal window on an Ubuntu system showing the output of the command 'sudo /var/ossec/bin/agent_control -l'. The output lists available agents. The second line, 'ID: 001, Name: windows-agent, IP: 192.168.56.103, Active', is highlighted with a red box.

```
root@ubuntu-VirtualBox:/home/ubuntu# sudo /var/ossec/bin/agent_control -l
Wazuh agent_control. List of available agents:
  ID: 000, Name: ubuntu-VirtualBox (server), IP: 127.0.0.1, Active/Local
  ID: 001, Name: windows-agent, IP: 192.168.56.103, Active
```

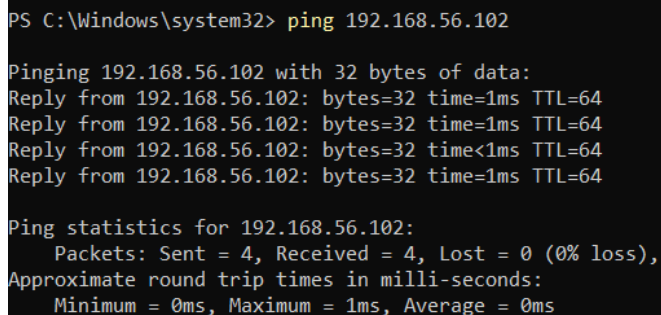
Figure 4.4: Verifying Agent Connection to the SOC Server

This confirms that the connection of the WindowsVM is active.

- b. On your host machine, first make sure your host machine can reach your Ubuntu VM IP

In powershell, on your host Windows, test:

```
ping 192.168.56.102
```

A Windows PowerShell terminal window showing the results of a ping command to 192.168.56.102. The output shows four successful replies with 32 bytes of data, a time of 1ms, and a TTL of 64. Ping statistics show 4 packets sent, 4 received, and 0% loss.

```
PS C:\Windows\system32> ping 192.168.56.102

Pinging 192.168.56.102 with 32 bytes of data:
Reply from 192.168.56.102: bytes=32 time=1ms TTL=64
Reply from 192.168.56.102: bytes=32 time=1ms TTL=64
Reply from 192.168.56.102: bytes=32 time<1ms TTL=64
Reply from 192.168.56.102: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 4.4.2: Checking network connection between SOC server (Ubuntu) and wazuh-agent (host-windows)

This confirms that your host machine can connect to the SOC server (Ubuntu) which is important for our lab. After the confirmation, run through every step like you did for your windows VM. The steps are identical, just install the agent, get a key, and import it but first, you will need to configure your firewall to allow the

connection if Windows blocks it. To configure your firewall, open *Windows Defender Firewall* on your computer and follow the prompt as shown in the image below;

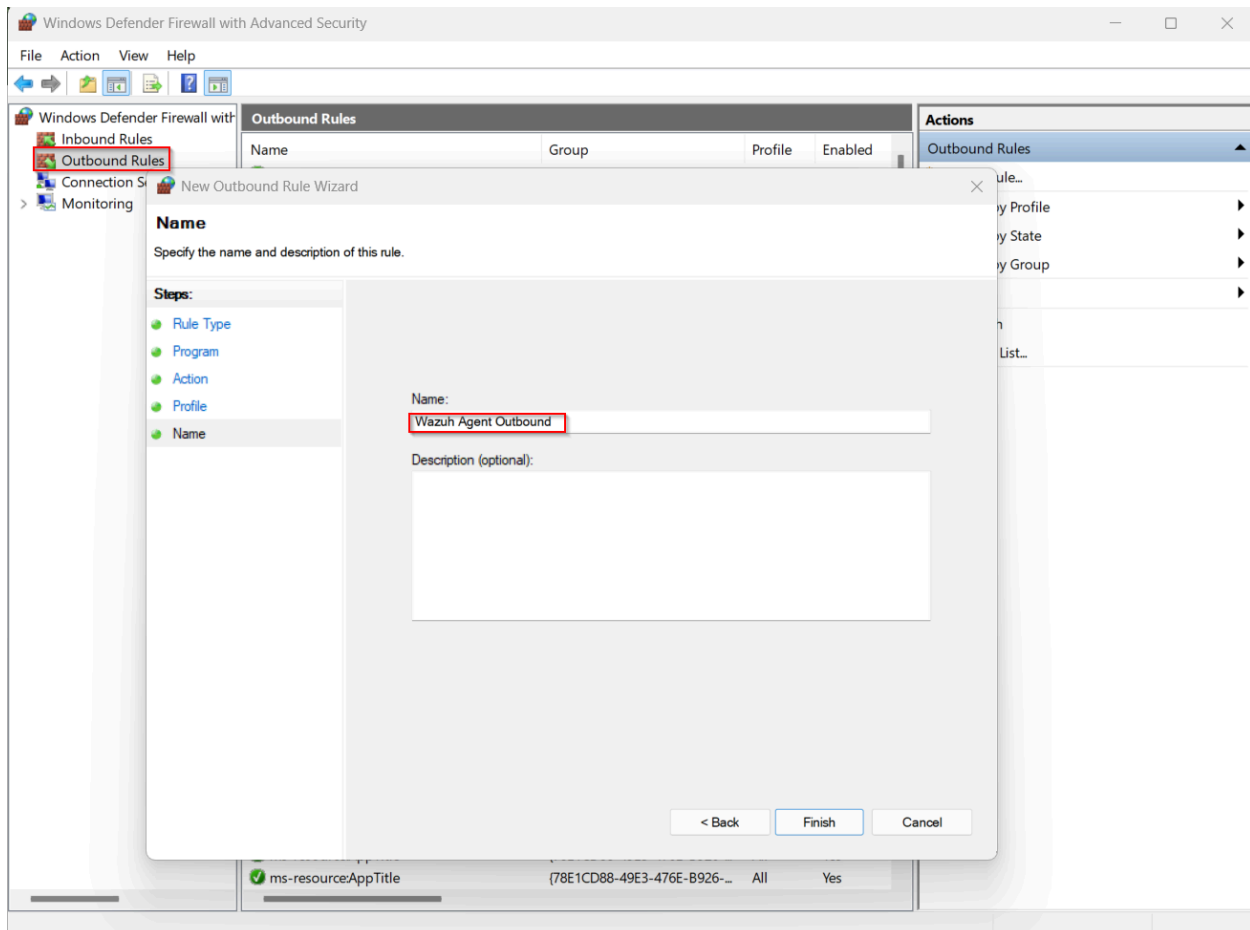


Figure 4.4.3: Configuring Windows Firewall to Allow Wazuh Agent Communication

After configuration, confirm in your SOC server (Ubuntu) if the agent (host-windows) has been connected successfully.

```
root@ubuntu-VirtualBox:/home/ubuntu# sudo /var/ossec/bin/agent_control -l

Wazuh agent_control. List of available agents:
ID: 000, Name: ubuntu-VirtualBox (server), IP: 127.0.0.1, Active/Local
ID: 001, Name: windows-agent, IP: 192.168.56.103, Active
ID: 003, Name: host-windows, IP: 192.168.56.1, Active
```

Figure 4.4.4: Confirming All Agents Are Connected to the SOC

Step 4.5: Check the Dashboard

Back in your Wazuh dashboard (the browser window), click **Agents** in the left menu. You should see all your connected machines with green checkmarks..

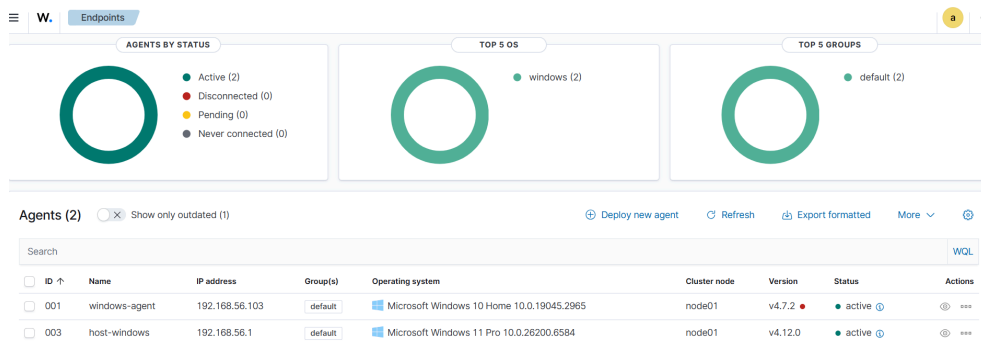


Figure 4.5: Connected Agents in the Wazuh Dashboard

Green means they are connected and sending data.

Part 5: Trigger Real Security Alerts

This is the part where we create alerts that trigger when suspicious activity happens. Normally, Wazuh comes with thousands of detection rules built in and that is why you may already see alerts coming in.

Test 1: Create a Custom Rule for Failed SSH Logins on Ubuntu

a. Add the rule

Let's create a rule that alerts us when someone tries to log into SSH multiple times:

- On your SOC server (Ubuntu) terminal, edit the rules file by using this command;

```
sudo nano /var/ossec/etc/rules/local_rules.xml
```

Add this rule:

```
<group name="ssh,">
  <rule id="100001" level="10">
```

```
<if_matched_sid>5710</if_matched_sid>
<description>SSH: Multiple failed login attempts detected
</description>
<group>authentication_failed,</group>
</rule>
</group>
```

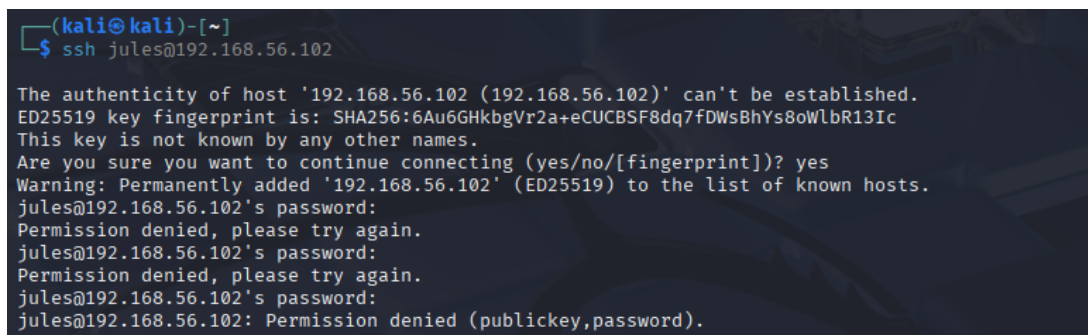
ii. Save (Ctrl+O and exit with Ctrl X) and restart Wazuh by typing the following command;

```
sudo systemctl restart wazuh-manager
```

b. Simulate an attack

From your Kali VM (attacker machine), try to SSH into the Ubuntu server with wrong passwords about 3 times:

```
ssh jules@192.168.56.102 # Your Ubuntu server IP
```



```
(kali@kali)-[~]
$ ssh jules@192.168.56.102

The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is: SHA256:6Au6GHkbgVr2a+eCUCBSF8dq7fDWsBhYs8oWlBR13Ic
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.102' (ED25519) to the list of known hosts.
jules@192.168.56.102's password:
Permission denied, please try again.
jules@192.168.56.102's password:
Permission denied, please try again.
jules@192.168.56.102's password:
jules@192.168.56.102: Permission denied (publickey,password).
```

Figure 5.1: Simulating an SSH Brute-Force Attack from Kali

c. See the alert

Back in Wazuh dashboard, go to *Security events* and search: *rule.id:100001*

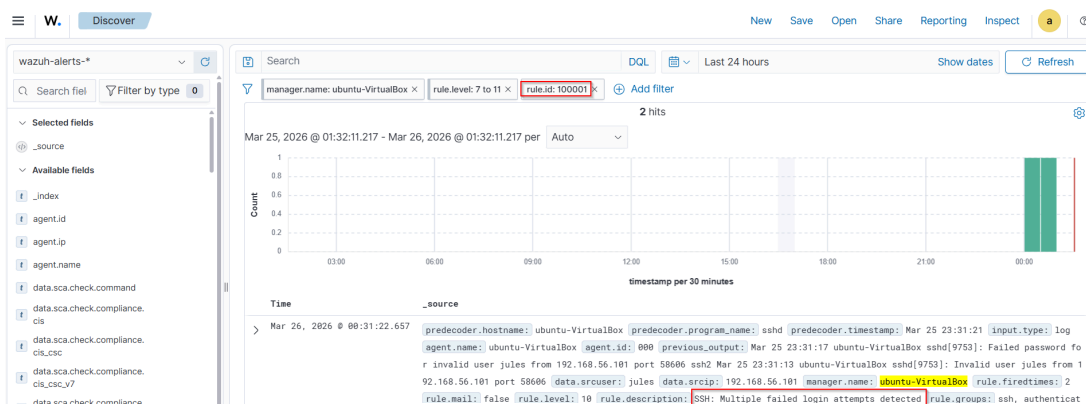


Figure 5.1.2: First Alert - SOC Detects the SSH Brute-Force Attempt

Now, let's break down this event in the best way possible.

- Timestamp: March 26, 2026 @ 00:31:22
- Source machine (attacker/attempting login): 192.168.56.101
- Target machine (SOC/Ubuntu agent): `ubuntu-VirtualBox`
- User tried to login as: `jules` (invalid user)

Key details from the log;

Field	Value / Example	Explanation
<code>predecoder.program_name</code>	sshd	Comes from the SSH daemon , meaning it's an SSH login attempt.
<code>Failed password for invalid user jules</code>	-	Someone tried to log in via SSH with wrong credentials for a user that doesn't exist on the server.
<code>rule.description</code>	SSH: Multiple failed login attempts detected	Wazuh triggered a security alert because the same source IP attempted to log in multiple times and failed.
<code>rule.firedtimes</code>	2	This is the second failed attempt that triggered the alert (part of repeated failures).
<code>rule.level</code>	10	Considered high severity in Wazuh.

Table 2: Breaking Down the SSH Brute-Force Alert

In summary, Someone from IP 192.168.56.101 (your Kali machine) tried to log into your Ubuntu server using a guessed user name (jules), and failed multiple times. Wazuh detected this as a potential brute-force attack and raised a level 10 alert.

d. What to do

In a real SOC environment, you should:

1. Confirm the IP is not legitimate (check if it's a known service)
2. Block the IP at your firewall
3. Check if any successful logins occurred from that IP
4. Review if that username exists (if it did, that user's account is compromised)

Test 2 - Create a User Account on Windows

For this, let us see how your SOC handles local account changes

a. Create a test user

On your Windows machine (main computer or Windows VM), open Command Prompt as Administrator and run:

```
netuserhacker123Password123!/add
```

```
PS C:\Windows\system32> net user hacker123 Password123! /add
The command completed successfully.
```

Figure 5.2: Creating a Test User Account on Windows

b. Find the alert

In your Wazuh dashboard, search for **hacker123**.

```
Mar 26, 2026 @ 02:09:46.844  input.type: log agent.ip: 192.168.56.1 agent.name: host-windows agent.id: 005 manager.name: ubuntu-VirtualBox
data.win.eventdata.subjectLogonId: 0x3c565 data.win.eventdata.scriptPath: %\1793 data.win.eventdata.passwordLastSet: 26/03/
2026 02:09:45 data.win.eventdata.homeDirectory: %\1793 data.win.eventdata.subjectDomainName: JULES
data.win.eventdata.displayName: %\1793 data.win.eventdata.accountExpires: %\1794 data.win.eventdata.homePath: %\1793
data.win.eventdata.samAccountName: hacker123 data.win.eventdata.targetUserName: hacker123
```

Figure 5.2.2: SOC Detects Account Creation Event (Event ID 4720)

Now, let's break down this event in the best way possible

Timestamp: March 26, 2026, 02:09:46

Agent: `host-windows` (IP: 192.168.56.1, Agent ID: 005)

SOC Manager: `ubuntu-VirtualBox`

Wazuh Level: 8 (high)

Event IDs: `4738` (Account changed), `4722` (Account enabled)

Key Details from the Log

Field	Value	Explanation
<code>subjectUserName</code>	THINKBOOK	The user who made the changes (administrator or logged-in user).
<code>targetUserName</code>	hacker123	The user account that was modified and enabled.
<code>oldUACValue</code>	0x15	Previous User Account Control settings.
<code>newUACValue</code>	0x10	New User Account Control settings.
<code>passwordLastSet</code>	26/03/2026 02:09:45	Timestamp when the account password was last set.
<code>accountExpires</code>	Never	Account is configured to never expire.
<code>primaryGroupId</code>	513	Default primary group ID for the account.
<code>MITRE Technique</code>	T1098 (Account Manipulation)	This indicates that the change could be used to maintain persistence in a system.
<code>rule.groups</code>	windows, windows_security, account_changed	Wazuh categorization of this event for monitoring purposes.

Table 3: Breaking Down the Windows Account Creation Alert

In summary, In Summary, A user account named "hacker123" was created, enabled, and had its properties modified. Wazuh flagged this because account changes, especially on Windows systems, are common tactics for attackers trying to maintain access or create backdoors.

d. What to do

In a real SOC environment, you should:

1. Verify who created the account (look at the "subjectUserName" field in the event)
2. Determine if the account should exist
3. Immediately disable suspicious accounts
4. Check recent login activity for that account
5. Review if the account was added to any privileged groups or if there were any other suspicious activity from that account.

Test 3: Delete That Same User Account

a. Delete the user

In the same Command Prompt, type `net user hacker123 /delete`

```
PS C:\Windows\system32> net user hacker123 /delete
The command completed successfully.
```

Figure 5.3: Deleting the Test Account to Generate the Removal Alert

b. Find the alert

Back in your dashboard, search again. You'll see **Event ID 4726: User account deleted**.

```
> Mar 26, 2026 @ 02:28:54.359 input.type: log agent.ip: 192.168.56.1 agent.name: host-windows agent.id: 005 manager.name: ubuntu-VirtualBox
data.win.eventdata.subjectLogonId: 0x3c565 data.win.eventdata.targetUserName: hacker123
data.win.eventdata.subjectUserSid: S-1-5-21-1632284433-856062334-1795990361-1001 data.win.eventdata.subjectDomainName: JULES
data.win.eventdata.targetDomainName: JULES data.win.eventdata.targetSid: S-1-5-21-1632284433-856062334-1795990361-1004
data.win.eventdata.subjectUserName: THINKBOOK data.win.system.eventID: 4726 data.win.system.keywords: 0x8020000000000000
```

Figure 5.3.2: SOC Detects Account Deletion Event (Event ID 4726)

Lets break down this event;

- Timestamp: Mar 26, 2026 @ 02:28:54
- Agent: `host-windows` (IP: 192.168.56.1)
- Manager: `ubuntu-VirtualBox` (SOC server)
- Event Type: Windows Security Event

Key Details from the Log

Field	Value	Explanation
<code>data.win.system.eventID</code>	4726	This is the Windows event ID for "User Account Deleted". It indicates a user account was removed from the system.
<code>data.win.eventdata.subjectUserName</code>	THINKBOOK	The account performing the action (who deleted the user).
<code>data.win.eventdata.subjectLogonId</code>	0x3c565	The logon session ID of the user performing the deletion.
<code>data.win.eventdata.subjectUserSid</code>	S-1-5-21-1632284433-856062334-1795990361-1001	The security identifier (SID) of the user performing the deletion.
<code>data.win.eventdata.targetUserName</code>	hacker123	The user account that was deleted.
<code>data.win.eventdata.targetSid</code>	S-1-5-21-1632284433-856062334-1795990361-1004	The SID of the deleted user account.
<code>data.win.eventdata.subjectDomainName</code> / <code>targetDomainName</code>	JULES	Both the account performing the action and the target account belong to the domain <code>JULES</code> .

Table 4: Breaking Down the Windows Account Deletion Alert

In summary, this event shows that the user *THINKBOOK* deleted the account *hacker123* on the Windows host. It is a critical action because deleting accounts can impact access control and persistence. In a SOC context, this is flagged under **user account management / security auditing**. In a real environment, this could be:

- A clean-up after legitimate testing
- An attacker covering their tracks

- Someone removing evidence of unauthorized access

The alert matters less in isolation. But combined with the creation event? That's a full audit trail of account activity.

Test 4: Vulnerability Detection (Critical Alert)

During testing, Wazuh detected a **critical vulnerability** on the Windows VM.

```
Mar 26, 2026 @ 21:27:59.165 input.type: log agent.ip: 192.168.56.103 agent.name: windows-agent agent.id: 001 manager.name: ubuntu-VirtualBox
data.vulnerability.severity: Critical data.vulnerability.package.condition: Package less than 10.0.19045.6575
data.vulnerability.package.name: Microsoft Windows 10 Home data.vulnerability.package.source: OS
data.vulnerability.package.version: 10.0.19045.6456 data.vulnerability.package.architecture: x86_64
data.vulnerability.assigner: microsoft data.vulnerability.cwe_reference: CWE-122 data.vulnerability.published: Nov 11, 2025 @
```

Figure 5.3.2: SOC Identifies CVE-2025-60724 on Windows VM

Unlike previous tests that detect attacks, this alert shows that your SOC can also identify weaknesses **before attackers exploit them**. This means your SOC is not just watching for someone trying to break in. It is actively scanning your systems, finding unlocked doors and open windows, and telling you about them before anyone else discovers them.

Key Details from the Log

Field	Value	What It Means
agent.name	windows-agent	This is your Windows VM, the affected machine
severity	Critical	Highest risk level
rule.level	13	Out of 15, very high alert in Wazuh
cve	CVE-2025-60724	The vulnerability's unique ID, like a serial number for this specific weakness
score.base	9.8 / 10	Extremely dangerous
attack_vector	NETWORK	Can be exploited remotely. The attacker does not need physical access.
privileges_required	NONE	Attacker needs no access to start exploiting this
user_interaction	NONE	No clicking, no opening files, or doing anything is required. Just being connected is enough.
impact	HIGH (C, I, A)	Affects Confidentiality, Integrity, Availability

Table 4: Breaking Down the Critical Vulnerability Alert (CVE-2025-60724)

In summary, the alert indicated that the system was running an outdated version of Windows 10 vulnerable to: **CVE-2025-60724 (CVSS score: 9.8)**. This vulnerability allows remote code execution without authentication or user interaction. In simpler words, it means that an attacker could gain full access remotely, execute malicious code and even deploy malware or ransomware, making it extremely dangerous.

b. What to do

In a real SOC environment, you should:

- Apply the latest Windows security update
- Patch the system to a secure version
- Check logs for suspicious activity around the time the vulnerability was present.

5.4: Repeated Failed Logon Detection (Authentication Alert)

Now let's look at something different, this is an alert that catches someone actively trying to break in.

During testing, after intentionally entering incorrect passwords during sign-in, Wazuh detected multiple failed logon attempts on the Windows VM.



Figure 5.4: Authentication Alert with Repeated Failed Logon Attempts Detected

This is different from vulnerability detection. Instead of finding weaknesses, this alert catches someone in the act of trying to exploit them.

```
> Mar 27, 2026 @ 04:44:30.498  input.type: log agent.ip: 192.168.56.103 agent.name: windows
                                -agent agent.id: 001 manager.name: ubuntu-VirtualBox
                                data.win.eventdata.subjectLogonId: 0x3e7
                                data.win.eventdata.subjectDomainName: WORKGROUP
                                data.win.eventdata.ipAddress: 127.0.0.1

> Mar 27, 2026 @ 04:44:27.326  input.type: log agent.ip: 192.168.56.103 agent.name: windows
                                -agent agent.id: 001 manager.name: ubuntu-VirtualBox
                                data.win.eventdata.subjectLogonId: 0x3e7
                                data.win.eventdata.subjectDomainName: WORKGROUP
                                data.win.eventdata.ipAddress: 127.0.0.1

> Mar 27, 2026 @ 04:44:24.433  input.type: log agent.ip: 192.168.56.103 agent.name: windows
                                -agent agent.id: 001 manager.name: ubuntu-VirtualBox
                                data.win.eventdata.subjectLogonId: 0x3e7
                                data.win.eventdata.subjectDomainName: WORKGROUP
                                data.win.eventdata.ipAddress: 127.0.0.1

> Mar 27, 2026 @ 04:44:21.105  input.type: log agent.ip: 192.168.56.103 agent.name: windows
                                -agent agent.id: 001 manager.name: ubuntu-VirtualBox
                                data.win.eventdata.subjectLogonId: 0x3e7
                                data.win.eventdata.subjectDomainName: WORKGROUP
```

Figure 5.4.2: Expanded Failed Logon Alert Event Details

Key Details from Log

Field	Value	What It Means
agent.name	windows-agent	The affected system is the Windows VM
eventID	4625	Windows event for "failed logon." This is the standard Windows audit event.
targetUserName	Jules	Account being targeted
ipAddress	127.0.0.1	The login attempt came from the machine itself, someone physically at the computer
logonType	2	Interactive logon, someone typing at the keyboard
processName	svchost.exe	System process handling authentication

Field	Value	What It Means
<code>status</code>	0xC000006D	Generic logon failure (username exists but password wrong)
<code>subStatus</code>	0xC000006A	Incorrect password
<code>rule.level</code>	5	Medium severity alert (something to watch, not an emergency)
<code>rule.firedtimes</code>	11	Alert triggered multiple times

Table 6: Understanding the Repeated Failed Logon Alert

Event Timeline Analysis

Look at how these events happened within a short timeframe:

Timestamp	What Happened	Analysis
04:44:21.105	First attempt	Initial authentication attempt
04:44:24.433	Second attempt	Repetition begins
04:44:27.326	Third attempt	Continuous failed attempts
04:44:30.498	Fourth attempt (fired times: 11)	Confirmed repeated activity

Table 7: Timeline of Repeated Authentication Failures

The logs show that the "Jules" account experienced multiple failed login attempts within approximately 13 seconds.

In summary, Someone tried to log into the "Jules" account. They failed. They tried again. Failed. Tried again. Failed. Eleven times total, all in under a minute. All attempts originated from the machine itself (127.0.0.1), used interactive logon (Logon Type 2), were processed by svchost.exe (system process) and failed due to incorrect password (0xC000006A) The increasing `firedtimes` value (up to 11) confirms that this was not a single failure, but a repeated sequence of authentication attempts. This pattern closely resembles a brute-force attack, where multiple passwords are tried in quick succession to gain access to an account.

b. What to do

In a real-world scenario, this would strongly suggest a brute-force attack, or an automated authentication attempt

1. Assess the pattern, check if they succeeded and identify the source
2. Configure Wazuh to alert sooner
3. Configure Windows to lock accounts after a certain number of failed attempts

Conclusion

A few hours ago, you would have said "I hope so" if someone asked if you would know your system was compromised.

Now you have a different answer. You built a real Security Operations Center. The same tools companies use to protect their infrastructure now run on your laptop, monitoring your machines. For free. You can now see failed logins in real time. Account changes. Critical vulnerabilities. Every time someone tries to get in.

That's the shift. Security is not about being invulnerable, it is about knowing when something goes wrong and responding before it becomes a disaster. Check your dashboard weekly. Add more devices. Let critical alerts find you.

Remember the 207-day statistic that I gave earlier? You just collapsed that to minutes. Most people never do this. They rely on luck. You built the tools to defend yourself and now you are prepared.